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Physical adjustment to system memory with chipset attached memory

Abstract

Physical adjustment to system memory with chipset attached memory is described. In accordance with the described techniques, an indication for making one or more physical adjustments to system memory of a device is received. Contents of the system memory are transferred via a chipset link to a chipset attached memory. The device is operated using the contents from the chipset attached memory while the one or more physical adjustments are made to adjust the system memory. After the one or more physical adjustments, the contents are transferred back from the chipset attached memory to the adjusted system memory via the chipset link.

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Background/Summary

BACKGROUND

(1) Computer games and other applications are regularly advancing, resulting in larger programs, higher resolution graphics, new features, and so forth. To address these advances, computer hardware is also advancing to provide new types of memory having faster data rates, faster clock rates, and so forth.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is an illustration of a non-limiting example system that employs a chipset attached memory to continue operation of the system during physical adjustment of system memory.

- (2) FIG. 2 depicts a non-limiting example in which chipset attached memory continues operation of a system during physical adjustment of system memory.
- (3) FIG. 3 depicts a procedure in an example implementation of physical adjustment to system memory with chipset attached memory.
- (4) FIG. 4 is an illustration of another non-limiting example system that employs a chipset attached memory to continue operation of the system during physical adjustment of system memory.
- (5) FIG. 5 is an illustration of another non-limiting example system that employs a chipset attached memory to continue operation of the system while the system memory is inoperable.

DETAILED DESCRIPTION

(6) Overview

(7) Physical adjustment to system memory with chipset attached memory is described. In accordance with the described techniques, a computing system uses a chipset attached memory to continue operating while adjustments, e.g., physical adjustments, are made to a system memory. For example, the system uses the chipset attached memory to continue running an operating system while the system memory is inoperable. To do so, the system memory is disabled (e.g., by gating the system memory so that the system memory is powered down and “offline”) which enables the system memory to be physically adjusted. Examples of such physical adjustments which can occur while the system memory is inoperable include, but are not limited to, removing, adding, or swapping one or more components of the system memory, such as a dual in-line memory module (DIMM), a small-outline dual in-line memory module (SO-DIMM), or a registered dual in-line memory module (RDIMM).

(8) In one or more implementations, contents of the system memory are transferred to the chipset attached memory. This transfer may occur, for example, responsive to a user request made via a user interface of the system. Once the contents are transferred to the chipset attached memory, various operations are performed using the contents which are now available for access in the chipset attached memory. For example, a processing unit of the system is able to execute (and/or continues to seamlessly execute) one or more tasks of an operating system and/or an application using the contents from the chipset attached memory. Notably, the processing unit uses the same contents that were in the main system memory but are now available transparently to the operating system or application in the chipset attached memory

(9) While such operations are performed using the contents from the chipset attached memory, the system memory is disabled and inoperable. As such, physical adjustments can be made to the system memory because the system memory is not in use, e.g., power to the system memory is gated or otherwise cut. Cutting power to the system memory (and other components), enables the system memory to be physically handled (e.g., touched) safely, such as by adding one or more components (e.g., a DIMM, SODIMM, or RDIMM) to the system memory, removing one or more components from the system memory, and/or swapping one or more components of the system memory. After physical adjustments are made to the system memory, the contents are transferred back from the chipset attached memory to the adjusted system memory. Once the contents are transferred back to the system memory, the processing unit executes (and/or continues executing) one or more tasks of an operating system and/or an application using the contents from the system memory.

(10) The ability to continue to operate by performing memory-based tasks while the system memory is inoperable and being physically adjusted contrasts with conventional techniques. For example, rather than continuing to run an operating system or an application on a processing unit while system memory is physically adjusted, conventional techniques also power down the processing unit (and/or the system altogether) so that the system memory can be physically adjusted in a safe manner. By continuing to operate using the chipset attached memory, however, the described techniques enable memory-based tasks to continue being performed during physical adjustments to the system memory. As a result, “down time” of the system is reduced and/or

eliminated, which reduces interruptions to various tasks (e.g., background tasks) and improves the user experience.

(11) In some aspects, the techniques described herein relate to an apparatus including: a system memory, a memory controller configured to provide a transfer request to the system memory to cause contents maintained by the system memory to be transferred from the system memory to a chipset attached memory, and a processing unit to execute one or more tasks using the contents from the chipset attached memory.

(12) In some aspects, the techniques described herein relate to an apparatus, further including a processing unit package that includes the processing unit and the memory controller, wherein the system memory is coupled to the processing unit package.

(13) In some aspects, the techniques described herein relate to an apparatus, further including a chipset link coupling the processing unit package to the chipset attached memory, wherein the contents is transferred from the system memory to the chipset attached memory via the chipset link.

(14) In some aspects, the techniques described herein relate to an apparatus, wherein the system memory is inoperable while the processing unit executes the one or more tasks using the contents from the chipset attached memory.

(15) In some aspects, the techniques described herein relate to an apparatus, wherein power to the system memory is gated while the processing unit executes the one or more tasks using the contents from the chipset attached memory.

(16) In some aspects, the techniques described herein relate to an apparatus, wherein gating the system memory enables one or more physical adjustments to be made to the system memory while the processing unit executes the one or more tasks using the contents from the chipset attached memory, wherein the one or more physical adjustments that can be made to the system memory include one or more of swapping a physical component of the system memory, removing a physical component of the physical memory, or replacing a physical component of the physical memory. 7.

(17) In some aspects, the techniques described herein relate to an apparatus, wherein the physical memory module includes a dual in-line memory module.

(18) In some aspects, the techniques described herein relate to an apparatus, wherein the memory controller is further configured to provide an additional transfer request to the chipset attached memory to cause the contents to be transferred back from the chipset attached memory to the system memory.

(19) In some aspects, the techniques described herein relate to an apparatus, wherein the memory controller is configured to provide the transfer request responsive to receiving an indication that one or more physical adjustments are going to be made to the system memory.

(20) In some aspects, the techniques described herein relate to a method including: receiving an indication for making one or more physical adjustments to system memory of a device, transferring, via a chipset link, contents of the system memory to a chipset attached memory, operating the device using the contents from the chipset attached memory while the one or more physical adjustments are made to adjust the system memory, and after the one or more physical adjustments, transferring, via the chipset link, the contents back from the chipset attached memory to the adjusted system memory.

(21) In some aspects, the techniques described herein relate to a method, further including cutting power to the system memory while the one or more physical adjustments are made to adjust the system memory.

(22) In some aspects, the techniques described herein relate to a method, further including operating the device using the contents from the adjusted system memory after the contents are transferred back from the chipset attached memory.

(23) In some aspects, the techniques described herein relate to a method, further including operating the device using the contents from the system memory before the contents are transferred to the chipset attached memory.

- (24) In some aspects, the techniques described herein relate to a method, wherein the one or more physical adjustments include one or more of swapping a physical component of the system memory, removing a physical component of the physical memory, or replacing a physical component of the physical memory.
- (25) In some aspects, the techniques described herein relate to a method, wherein the physical component includes a physical memory module.
- (26) In some aspects, the techniques described herein relate to a method, wherein the physical memory module includes a dual in-line memory module.
- (27) In some aspects, the techniques described herein relate to a method, further including training the adjusted physical memory before transferring the contents back from the chipset attached memory to the adjusted system memory.
- (28) In some aspects, the techniques described herein relate to a system including: system memory slots into which system memory is installable, a chipset attached memory, a memory controller configured to cause contents stored in the system memory to be transferred to the chipset attached memory, and a processing unit configured to execute one or more tasks using the contents from the chipset attached memory while one or more system memory components are added or removed from one or more of the system memory slots.
- (29) In some aspects, the techniques described herein relate to a system, wherein the one or more system memory components include one or more physical memory modules.
- (30) In some aspects, the techniques described herein relate to a system, wherein the one or more system memory components include one or more dual in-line memory modules.
- (31) FIG. 1 is an illustration of a non-limiting example system **100** that employs a chipset attached memory to continue operation of the system during physical adjustment of system memory. The system **100** includes a processing unit package **102**, a system memory **104**, an I/O expander **106**, a chipset attached memory **108**, and a chipset link **110**. The processing unit package **102**, the system memory **104**, the I/O expander **106**, the chipset attached memory **108**, and the chipset link **110** are installed on or are part of, for example, a motherboard or other printed circuit board. In one or more implementations, the I/O expander **106** (including the chipset attached memory controller **116**), the chipset link **110**, one or more memory channels between the processing unit package **102** and the system memory **104**, and one or more memory channels between the I/O expander **106** and the chipset attached memory **108** are also referred to as a chipset of a motherboard or other printed circuit board. In one or more variations, the chipset link **110** could be routed entirely on the motherboard itself. In one or more other variation, the I/O expander **106** (including the chipset attached memory controller **116**) and the chipset attached memory **108** could be implemented on an add-in card that is connected to the motherboard via a socket.
- (32) The processing unit package **102** includes a processing unit **112** and a memory controller **114**. The processing unit **112** is any of various processing units, such as a central processing unit (CPU), a graphics processing unit (GPU), an Accelerated Processing Unit (APU), a parallel accelerated processor, a digital signal processor, an artificial intelligence (AI) or machine learning accelerator, and so forth. Although a single processing unit **112** is illustrated in the system **100**, the processing unit package **102** optionally includes any number of processing units of the same or different types.
- (33) The system memory **104** is any of a variety of types of physical RAM. Examples of system memory **104** include dynamic random-access memory (DRAM), phase-change memory (PCM), memristors, static random-access memory (SRAM), and so forth. The system memory **104** is coupled or attached to the processing unit package **102** via one or more memory channels. The system memory **104** is packaged or configured in any of a variety of different manners. Examples of such packaging or configuring include a dual in-line memory module (DIMM), a small-outline DIMM (SO-DIMM), a registered dual in-line memory module (RDIMM), a non-volatile DIMM (NVDIMM), a ball grid array (BGA) memory permanently attached to (e.g., soldered to) the motherboard (or other printed circuit board), and so forth.

(34) Examples of types of DIMMs include, but are not limited to, synchronous dynamic random-access memory (SDRAM), double data rate (DDR) SDRAM, double data rate 2 (DDR2) SDRAM, double data rate 3 (DDR3) SDRAM, double data rate 4 (DDR4) SDRAM, and double data rate 5 (DDR5) SDRAM. In at least one variation, the system memory **104** is configured as or includes a SO-DIMM or an RDIMM according to one of the above-mentioned standards, e.g., DDR, DDR2, DDR3, DDR4, and DDR5.

(35) Further examples of memory configurations include low-power double data rate (LPDDR), also known as LPDDR SDRAM, which is a type of synchronous dynamic random-access memory. In variations, LPDDR consumes less power than other types of memory and/or has a form factor suitable for mobile computers and devices, such as mobile phones. Examples of LPDDR include, but are not limited to, low-power double data rate 2 (LPDDR2), low-power double data rate 3 (LPDDR3), low-power double data rate 4 (LPDDR4), and low-power double data rate 5 (LPDDR5). It is to be appreciated that the System memory **104** is configurable in a variety of ways without departing from the spirit or scope of the described techniques.

(36) The memory controller **114** manages access to the system memory **104**, such as by sending read and write requests to the system memory **104** and receiving responses from the system memory **104**. In one or more implementations, the system memory **104** is physical memory managed by an operating system running on the processing unit **112** (e.g., a CPU), such as by allocating portions of the system memory **104** to applications running on the processing unit **112**, managing virtual memory spaces and memory pages for applications running on the processing unit **112**, and so forth.

(37) The processing unit package **102** optionally includes one or more additional controllers to link to additional devices, such as a Peripheral Component Interconnect Express (PCIe) controller, a Serial Advanced Technology Attachment (SATA) controller, a Universal Serial Bus (USB) controller, a Serial Peripheral Interface (SPI) controller, a Low Pin Count (LPC) controller, and so forth. Additionally or alternatively, one or more of these additional controllers is implemented separate from the processing unit package **102**, such as in a chip (e.g., an integrated circuit optionally referred to as a northbridge) that is part of the chipset of a motherboard or other printed circuit board.

(38) The processing unit package **102** communicates with the I/O expander **106** via the chipset link **110**. The chipset link **110** is any of a variety of communication links, such as a high-speed bus. In one example, the chipset link **110** is one or more PCIe lanes.

(39) The I/O expander **106** includes a chipset attached memory controller **116**. The I/O expander **106** optionally includes or is coupled to one or more additional controllers to link to other devices, such as a PCIe controller, a SATA controller, a USB controller, an SPI controller, an LPC controller, and so forth. In one or more implementations, the I/O expander **106** is referred to as a southbridge.

(40) The chipset attached memory controller **116** manages access to the chipset attached memory **108**, such as by sending read and write requests to the chipset attached memory **108** and receiving responses from the chipset attached memory **108**. The chipset attached memory **108** is referred to as “chipset attached” due to the chipset attached memory **108** being attached to the I/O expander **106** rather than the processing unit package **102** directly, and due to the chipset attached memory **108** being controlled by a memory controller of the I/O expander **106** rather than a memory controller of the processing unit package **102**. The chipset attached memory **108** is coupled or attached to the I/O expander **106** via one or more memory channels.

(41) The chipset attached memory **108** is physical memory managed by an application or program other than an operating system running on the processing unit **112**. The chipset attached memory **108** is separate from the system memory **104**, allowing the chipset attached memory to be used in various manners, such as to speed up access to frequently used data, without reducing the amount of system memory **104** available to an operating system running on the processing unit **112**.

(42) The chipset attached memory **108** is any of a variety of types of physical memory. Examples of chipset attached memory **108** include random-access memory (RAM), such as DRAM, PCM, memristors, SRAM, and so forth. The chipset attached memory **108** is volatile memory or nonvolatile memory. The chipset attached memory **108** is packaged or configured in any of a variety of different manners. Examples of such packaging or configuring include a DIMM, a SO-DIMM, an RDIMM, an NVDIMM, a BGA, a 3-dimensional (3D) stacked memory, on-package memory (e.g., memory included in the I/O expander **106**), memory permanently attached to (e.g., soldered to) the motherboard, and so forth.

(43) As noted above, examples of types of DIMMs include, but are not limited to, SDRAM, DDR SDRAM, DDR2 SDRAM, DDR3 SDRAM, DDR4 SDRAM, and DDR5 SDRAM. In at least one variation, the chipset attached memory **108** is configured as or includes a SO-DIMM or an RDIMM according to one of the above-mentioned standards, e.g., DDR, DDR2, DDR3, DDR4, and DDR5. Further examples of chipset attached memory configurations include LPDDR, such as LPDDR2, LPDDR3, LPDDR4, and LPDDR5. It is to be appreciated that the chipset attached memory **108** is configurable in a variety of ways without departing from the spirit or scope of the described techniques.

(44) In accordance with the described techniques, the system **100** uses the chipset attached memory **108** to continue operating while adjustments, e.g., physical adjustments, are made to the system memory **104**. For example, the system **100** uses the chipset attached memory **108** to continue running an operating system while the system memory **104** is being physically adjusted and is thus powered down and “offline.” Examples of such physical adjustments include, but are not limited to, removing the system memory **104** (or one or more components of the system memory **104**, such as a DIMM, a SODIMM, or an RDIMM), swapping the system memory **104** (or components of the system memory **104**, such as DIMMs, SODIMMs, or RDIMMs), replacing the system memory **104** (or one or more components of the system memory **104**, such as a DIMM, a SODIMM, or an RDIMM), and adding to the system memory **104** (e.g., adding one or more components, such as a DIMM, SODIMM, or RDIMM).

(45) In one or more implementations, an indication **118** for making physical adjustments to the system memory **104** is received. By way of example, a user provides an input via a user interface of a system- or memory-management application (not shown), being executed by the processing unit **112**, and the input indicates that the user intends to physically adjust the system memory **104**. This enables the system **100** to transition to a state where a person can physically adjust the system memory **104** without endangering himself or herself and without damaging components of the system **100**. If power is not cut to the system memory **104**, for instance, a person attempting to physically adjust the system memory **104** risks being electrically shocked, which can cause a variety of adverse outcomes and can be fatal in some circumstances.

(46) In the illustrated example, the memory controller **114** is depicted receiving the indication **118** for making the physical adjustments. In one or more implementations, the indication **118** is received from an operating system and/or an application, e.g., executing on the processing unit **112** or portions of which execute on the processing unit **112**. In at least one variation, the indication **118** is received from another component of the system **100** or from a different component (not shown) coupled to the system **100**. Additionally, in one or more scenarios, the indication **118** is not received based on user input—the indication **118** is received based on a different trigger.

(47) In accordance with the described techniques, contents **120** of the system memory **104** are transferred via the chipset link **110** to the chipset attached memory **108**. Based on or responsive to receipt of the indication **118**, for instance, the memory controller **114** causes the contents **120** of the system memory **104** to be transferred via the chipset link **110** to the chipset attached memory **108**.

(48) Once the contents **120** are transferred to the chipset attached memory **108**, various operations are performed using the contents **120** from the chipset attached memory **108**. For example, the processing unit **112** executes (and/or continues to execute) one or more tasks of an operating

system and/or an application using the contents **120** from the chipset attached memory **108**. Such tasks are performed using the contents **120** from the chipset attached memory **108** rather than using the contents **120** from the system memory **104**. While such operations are performed using the contents **120** from the chipset attached memory **108**, physical adjustments can be made to the system memory **104** because the system memory **104** is not in use, e.g., power to the system memory **104** is gated or otherwise cut. In one or more implementations, one or more components of the system **100** perform one or more actions in addition to cutting power to the system memory **104** to enable safe physical modification of the system memory **104** and/or portions of the system memory. By cutting power to the system memory **104** (and other components), the system **100** enables the system memory **104** to be physically handled (e.g., touched) safely, so that one or more components (e.g., a DIMM, SODIMM, or RDIMM) can be added to the system memory **104**, one or more components can be removed from the system memory **104**, and/or one or more components of the system memory **104** can be swapped.

(49) Continuing to operate by performing memory-based tasks while the system memory **104** is being physically adjusted, contrasts with conventional techniques. Rather than continuing to run an operating system or an application on a processing unit while system memory is physically adjusted, conventional techniques also power down the processing unit (and/or the system **100** altogether) so that the system memory can be physically adjusted in a safe manner. By continuing to operate using the chipset attached memory **108**, the described techniques enable memory-based tasks to continue being performed (which keeps operating systems, applications, and so on running) during physical adjustments to the system memory **104**. As a result, “down time” of the system **100** is reduced and/or eliminated, which reduces interruptions to various tasks (e.g., background tasks) and improves a user experience.

(50) After physical adjustments are made to the system memory **104**, the contents **120** are transferred back from the chipset attached memory **108** to the system memory **104** via the system memory **104**, which has been adjusted. In one or more implementations, prior to transferring the contents **120** back to the system memory **104**, the new physical configuration of the system memory (e.g., with new DIMMs) is trained and brought back online and then the active contents **120** in the chipset attached memory **108** are transferred back to the system memory **104**. Once the contents **120** are transferred back to the system memory **104**, various operations are performed using the contents **120** from the system memory **104** as adjusted—rather than performing the operations using the contents **120** from the chipset attached memory **108**. For example, the processing unit **112** executes (and/or resumes executing) one or more tasks of an operating system and/or an application using the contents **120** from the system memory **104**.

(51) Thus, in one or more scenarios, the processing unit **112** operates using the contents **120** from the system memory **104**, e.g., before the contents **120** are transferred to the chipset attached memory **108**. The contents **120** are then transferred from the system memory **104** to the chipset attached memory **108** via the chipset link **110**. The processing unit **112** subsequently operates (or continues to operate) using the contents **120** from the chipset attached memory **108**, e.g., while the system memory **104** is being physically adjusted. The contents **120** are transferred from the chipset attached memory **108** to the system memory **104** via the chipset link **110**, e.g., after the physical adjustments to the system memory **104**. The processing unit **112** subsequently operates (or continues to operate) using the contents **120** from the system memory **104**.

(52) In one or more implementations, the system **100** outputs one or more indications when it is safe for a user to physically adjust the system memory **104**. After the contents **120** are transferred to the chipset attached memory **108** and operation of the system **100** continues using the contents **120** from the chipset attached memory **108**, for instance, the system **100** may output one or more visual or haptic indicators. By way of example, and not limitation, the system **100** may output a signal (e.g., flash, change color, strobe, turn on, turn off, etc.) via a light emitting diode (LED) on a component (e.g., the processing unit package **102** or near a socket for the system memory **104**),

output one or more sounds via a speaker (e.g., beeps, chimes, etc.), output a haptic signal (e.g., a vibration) via one or more components, and so on. Alternately or in addition, a system- or memory-management application (not shown) operating on the processing unit **112** outputs a notification via a user interface indicating that it is safe for a user to physically adjust the system memory **104**, which is possible because the operations of the system are continued using the contents **120** from the chipset attached memory **108**. In one or more implementations, a notification may be output via one or more devices connected to the system **100**. For example, the notification may be communicated from the system **100** to the device via a wireless connection. By way of example and not limitation, the notification may be communicated from the system **100** to the device via a Bluetooth connection, via a Wi-Fi connection, or via the Internet (e.g., as an email message). The device connected to the system may then output the notification, e.g., via a user interface of an application associated with the system **100**. One or more of these various indications inform a user when it is safe to physically adjust the system memory **104**, and thereby reducing or eliminating adverse outcomes that can result from handling system memory at a wrong time.

(53) The system **100** is implementable in a device or apparatus. Examples of the different types of devices or apparatuses in which the system **100** can be implemented include, by way of example and not limitation, a personal computer (e.g., a desktop or tower computer), a smartphone or other wireless phone, a tablet or phablet computer, a notebook computer, a laptop computer, a wearable device (e.g., a smartwatch, an augmented reality headset or device, a virtual reality headset or device), an entertainment device (e.g., a gaming console, a portable gaming device, a streaming media player, a digital video recorder, a music or other audio playback device, a television, a set-top box), an Internet of Things (IoT) device, an automotive computer, and so forth.

(54) FIG. 2 depicts a non-limiting example **200** in which chipset attached memory continues operation of a system during physical adjustment of system memory. The example **200** includes the system memory **104**, the memory controller **114**, the chipset link **110**, the chipset attached memory controller **116**, and the chipset attached memory **108**.

(55) The example **200** includes a variety of example communications and operations between the system memory **104**, the memory controller **114**, the chipset link **110**, the chipset attached memory controller **116**, and the chipset attached memory **108** over time. In this example **200**, the communications and operations are positioned vertically based on time, such that communications and operations closer to a top of the example occur prior to communications or operations further from the top of the example. It follows also that communications or operations closer to a bottom of the example occur subsequent to communications or operations further from the bottom. The example **200** also depicts various phases and/or states of the system **100** or portions of the system **100**. These phases and/or states are also positioned in the example **200** vertically based on time, such that phases or states closer to a top of the example occur prior to phases, states, or communications further from the top.

(56) Here, the illustrated example **200** depicts the system memory **104** receiving one or more access requests **202(a)** (e.g., read and/or write requests) from the memory controller **114**. The illustrated example **200** also depicts the memory controller **114** receiving one or more serviced requests **204(a)** from the system memory **104**. Where an access request **202(a)** corresponds to a read request, for instance, the respective serviced request **204(a)** includes or otherwise indicates data of one or more memory addresses associated with the access request **202(a)**. In contrast, where an access request **202(a)** corresponds to a write request, the respective serviced request **204(a)** involves updating the system memory **104** at one or more memory addresses associated with the write request to maintain one or more indicated values.

(57) Transmission of the access requests **202(a)** and the servicing of them (e.g., as indicated by the serviced requests **204(a)**) by the memory controller **114** and the system memory **104** represents a first phase **206** of operation of a device (and/or the system **100**), which uses the contents **120** from the system memory **104**. Transmission of the access requests **202(b)** and the servicing of them (e.g.,

as indicated by the serviced requests **204(b)**) between the memory controller **114**, the chipset attached memory controller **116**, and the chipset attached memory **108** represents a second phase **208** of operation of the device (and/or the system **100**), which uses the contents **120** from the chipset attached memory **108**. The subsequent transmission of the access requests **202(c)** and the servicing of them (e.g., as indicated by the serviced requests **204(c)**) by the memory controller **114** and the system memory **104** represents a third phase **210** of operation of the device (and/or the system **100**), which uses the contents **120** from the system memory **104**.

(58) In this example **200**, the memory controller **114** is depicted receiving the indication **118** at a point in time during the first phase **206** of operation. It is to be appreciated that, in various scenarios, the indication **118** is received interleaved with transmission of one or more access requests **202(a)** and indications of serviced requests **204(a)**. As noted above, in one or more implementations, the indication **118** is received based on user input, e.g., received via a user interface and indicating that physical adjustments are going to be made to the system memory **104**.

(59) In one or more implementations, the memory controller **114** provides a transfer request **212** to the system memory **104** (e.g., one or more write requests), e.g., for transferring contents **120** maintained by the system memory **104** to the chipset attached memory **108**. In variations, the contents **120** transferred ranges from a subset of data maintained in the system memory **104** to an entirety of the data maintained in the system memory **104**. In terms of data flow, the contents **120** are transferred from the system memory **104** (e.g., over one or more memory interfaces) to the memory controller **114**, which communicates the contents **120** over the chipset link **110** to the chipset attached memory controller **116** (e.g., of the I/O expander **106**), and the chipset attached memory controller **116** communicates the contents **120** to the chipset attached memory **108** (e.g., over one or more memory interfaces), where the contents are stored. In one or more implementations, the chipset attached memory controller **116** stores the contents **120** in the chipset attached memory **108** using one or more write requests.

(60) After the contents are transferred, operation of the device (and/or the system **100**) uses the contents **120** from the chipset attached memory **108**. This example **200** depicts overlap **214** between when chipset attached memory **108** includes the contents **120**, such that the system **100** is operable using the contents in the chipset attached memory **108**, and when the system **100** is still operable using the system memory **104**. In one or more variations, any of one or more components of the system **100** is configured to test the chipset attached memory **108** during such an overlap, e.g., to ensure that the chipset attached memory **108** is capable of operating properly when the system memory **104** is powered down and/or otherwise taken offline. However, in at least one other variation, the chipset attached memory **108** is not tested before powering down and/or otherwise taking the system memory **104** offline.

(61) In the illustrated example **200**, operability of the system memory **104** is indicated by black bars arranged perpendicularly to the system memory **104** block whereas non-operability or a powered-down state is indicated by the white bar arranged between the black bars and perpendicular to the system memory **104** block. Similarly, operability of the chipset attached memory **108** is indicated by the black bar arranged perpendicularly to the chipset attached memory **108** block that is disposed on the dashed line extending from the chipset attached memory **108** block.

(62) During the second phase **208** of operation, the device (and or the system **100**) operates using the contents **120** from the chipset attached memory **108**, e.g., rather than using the system memory **104**. Thus, the chipset attached memory **108** services read/write requests previously serviced by the system memory **104** during the first phase **206** of operation. In other words, during the second phase **208**, the chipset attached memory **108** services one or more of the access requests **202(b)**, resulting in serviced requests **204(b)** while the system memory **104** is non-operable. In this example **200**, the subsequent access requests **202(b)** are depicted as being communicated from the memory controller **114** over the chipset link **110** to the chipset attached memory controller **116** and

then to the chipset attached memory **108**. By way of example, the access requests **202(b)** are initiated by the processing unit **112**, e.g., for executing one or more tasks of an operating system and/or application. Further, the serviced requests **204(b)** are depicted being provided from the chipset attached memory **108** to the chipset attached memory controller **116** and over the chipset link **110** to the memory controller **114**.

(63) During the second phase **208** of operation, the device is operated using the contents **120** from the chipset attached memory **108** and the system memory **104** is inoperable. As discussed throughout, during this period in which the system memory **104** is inoperable, one or more physical adjustments **216** can be made to the system memory **104**. In the illustrated example **200**, the physical adjustments **216** are indicated being made while the system memory **104** is powered down or otherwise non-operable, e.g., as indicated by the white bar perpendicularly arranged below the system memory **104** box. Examples of physical adjustments made to the system memory **104** during this time interval are described in more detail above. In one or more implementations, the system memory **104** can be rendered inoperable for other reasons. For example, the system memory **104** may be rendered inoperable due to a system failure or corruption. In these scenarios, the contents **120** can be transferred from the system memory **104** to the chipset attached memory **108** while a system level reboot is performed.

(64) After the physical adjustments, the contents **120** are transferred back to the system memory **104** from the chipset attached memory **108** via the chipset link **110**. In one or more implementations, the system memory **104**, as adjusted, is trained before the contents **120** are transferred back to the system memory **104**. In one or more implementations, the chipset attached memory controller **116** receives an indication (not shown) that indicates the system memory **104** is again operable, and is thus capable of servicing read/write requests. In at least one variation, such an indication indicates to the chipset attached memory controller **116** to initiate transfer of the contents **120** back to the adjusted system memory **104**. In variations, the contents **120** transferred ranges from a subset of data maintained in the system memory **104** to an entirety of the data maintained in the system memory **104**. In terms of data flow, the contents **120** are transferred from the chipset attached memory **108** (e.g., over one or more memory interfaces) to the chipset attached memory controller **116**, which communicates the contents **120** over the chipset link **110** to the memory controller **114** (e.g., of the processing unit package **102**), and the memory controller **114** communicates the contents **120** to the system memory **104** (e.g., over one or more memory interfaces), where the contents **120** are stored. In one or more implementations, the memory controller **114** stores the contents **120** in the system memory **104** using one or more write requests.

(65) After the contents are transferred, operation of the device (and/or the system **100**) uses the contents **120** from the system memory **104**. This example **200** depicts overlap **218** between when system memory **104** includes the contents **120**, such that the system **100** is operable using the contents in the system memory **104**, and when the system **100** is still operable using the chipset attached memory **108**. In one or more variations, any of one or more components of the system **100** is configured to test or train the system memory **104** during such an overlap, e.g., to ensure that the system memory **104** is capable of operating properly when operation of the system returns to using the system memory **104** instead of using the chipset attached memory **108**. However, in at least one other variation, the system memory **104** is not tested before returning operation from the chipset attached memory **108** to the system memory **104**.

(66) During the third phase **210** of operation, the device (and or the system **100**) operates using the contents **120** from the system memory **104** as physically adjusted, e.g., rather than using the chipset attached memory **108**. Thus, the adjusted system memory **104** services read/write requests, which were serviced by the chipset attached memory **108** during the second phase **208** of operation. In other words, during the third phase **210**, the system memory **104** again services one or more of the access requests **202(c)**, resulting in serviced requests **204(c)**.

(67) FIG. 3 depicts a procedure in an example **300** implementation of physical adjustment to

system memory with chipset attached memory.

(68) An indication for making one or more physical adjustments to system memory of a device is received (block **302**). By way of example, an indication **118** for making physical adjustments to the system memory **104** is received. In one or more implementations, the indication **118** is received responsive to user input provided via a user interface of a system- or memory-management application (not shown), being executed by the processing unit **112**, and the input indicates that the user intends to physically adjust the system memory **104**.

(69) Contents of the system memory are transferred to a chipset attached memory via a chipset link (block **304**). By way of example, contents **120** of the system memory **104** are transferred via the chipset link **110** to the chipset attached memory **108**. Based on or responsive to receipt of the indication **118**, for instance, the memory controller **114** causes the contents **120** of the system memory **104** to be transferred via the chipset link **110** to the chipset attached memory **108**.

(70) The device is operated using the contents from the chipset attached memory while the one or more physical adjustments are made to adjust the system memory (block **306**). By way of example, once the contents **120** are transferred to the chipset attached memory **108**, various operations are performed using the contents **120** from the chipset attached memory **108**. For example, the processing unit **112** executes (and/or continues to execute) one or more tasks of an operating system and/or an application using the contents **120** from the chipset attached memory **108**. Such tasks are performed using the contents **120** from the chipset attached memory **108** rather than using the contents **120** from the system memory **104**. While such operations are performed using the contents **120** from the chipset attached memory **108**, physical adjustments can be made to the system memory **104** because the system memory **104** is not in use, e.g., power to the system memory **104** is gated or otherwise cut.

(71) After the one or more physical adjustments, the contents are transferred back from the chipset attached memory to the adjusted system memory via the chipset link (block **308**). By way of example, after physical adjustments are made to the system memory **104**, the contents **120** are transferred back from the chipset attached memory **108** to the system memory **104** via the system memory **104**, which has been adjusted. Once the contents **120** are transferred back to the system memory **104**, various operations are performed using the contents **120** from the system memory **104** as adjusted—rather than performing the operations using the contents **120** from the chipset attached memory **108**. For example, the processing unit **112** executes (and/or resumes executing) one or more tasks of an operating system and/or an application using the contents **120** from the system memory **104**.

(72) FIG. 4 is an illustration of another non-limiting example system **400** that employs a chipset attached memory to continue operation of the system during physical adjustment of system memory. The system **400** includes the processing unit package **102**, the system memory **104**, the I/O expander **106**, the chipset attached memory **108**, and the chipset link **110** analogous to the system **100** of FIG. 1. The system **400** also includes a network I/O controller **402** and a chipset link **404** allowing the processing unit package **102** to communicate with the network I/O controller **402** using any of a variety of communication links analogous to chipset link **110**.

(73) The processing unit package **102**, the system memory **104**, the I/O expander **106**, the chipset attached memory **108**, the chipset link **110**, the network I/O controller **402**, and the chipset link **404** are installed on or are part of, for example, a motherboard or other printed circuit board. In one or more implementations, the I/O expander **106** (including the chipset attached memory controller **116**), the chipset link **110**, the chipset link **404**, one or more memory channels between the processing unit package **102** and the system memory **104**, and one or more memory channels between the I/O expander **106** and the chipset attached memory **108** are also referred to as a chipset of a motherboard or other printed circuit board.

(74) The processing unit package **102** includes the processing unit **112** and the memory controller **114**. Although a single processing unit **112** is illustrated in the system **400**, the processing unit

package **102** optionally includes any number of processing units of the same or different types, and or other types of components, such as an artificial intelligence accelerator. Given this architecture, such other optional components also access the system memory **104** directly (e.g., via the memory controller **114**) or through an operating system running on the processing unit **112**. Those components are also configured to access the chipset attached memory **108** (e.g., via the I/O expander **106** or the network I/O controller **402**).

(75) The network I/O controller **402** manages communication over a network, such as by sending data or control signals to one or more other devices via the network and receiving data or control signals from one or more other devices via the network. The network is implemented in any of a variety of manners, such as an Ethernet network, an InfiniBand network, and so forth. The network I/O controller **402** is also coupled or attached to the chipset attached memory **108** via one or more memory channels. In one or more implementations, the chipset attached memory **108** is address space (e.g., PCIe address space) that is addressable by other server nodes as well as components of the system **400** (e.g., the processing unit **112**). The network I/O controller **402** is thus able to send read and write requests to the chipset attached memory **108** and receive responses from the chipset attached memory **108**, analogous to the chipset attached memory controller **116**.

(76) In one or more implementations the chipset attached memory **108** is attached or coupled to only one of the network I/O controller **402** or the I/O expander **106** rather than attached or coupled to both the network I/O controller **402** and the I/O expander **106**.

(77) The network I/O controller **402** being attached or coupled to the chipset attached memory **108** supports various different usage scenarios. In one or more implementations, the processing unit package **102** is able to access the chipset attached memory **108** via the network I/O controller **402**, allowing the chipset attached memory **108** to be used in situations where the chipset attached memory **108** is not attached or coupled to the I/O expander **106**. Such situations arise, for example, where board routing limitations prevent the chipset attached memory **108** from being attached or coupled to the I/O expander **106**.

(78) In one or more implementations, the network I/O controller **402** allows the chipset attached memory **108** to be accessed by other devices via the network. This access is allowed using any of a variety of public or proprietary remote direct memory access (RDMA) techniques. For example, assume the system **400** is implemented in a server node connected to multiple other server nodes (e.g., some including their own chipset attached memory and optionally others not including their own chipset attached memory). Another server node communicates read and write requests to the chipset attached memory **108** via the network I/O controller **402** and receives responses from the chipset attached memory **108** via the network I/O controller **402**. The other server node is thus able to make use of the chipset attached memory **108** without disrupting the system memory **104** or even the processing unit package **102**. E.g., the processing unit package **102** need not have knowledge of the other server node accessing the chipset attached memory **108**.

(79) By way of another example, assume the system **400** is implemented in a server node connected to multiple other server nodes, at least one of which includes its own chipset attached memory. In at least one variation, the processing unit **112** (or other component of the processing unit package **102**) is able to communicate read and write requests to the chipset attached memory of the other server node via the network I/O controller **402** and receive responses from the chipset attached memory of the other server node via the network I/O controller **402**. The processing unit **112** or other component of the processing unit package **102** is thus able to make use of the chipset attached memory of another sever node without disrupting the system memory **104** or the chipset attached memory **108**.

(80) FIG. 5 is an illustration of another non-limiting example system **500** that employs a chipset attached memory to continue operation of the system while the system memory is inoperable. The system **500** is, for example, a motherboard or other printed circuit board. The system **500** includes an I/O expander **502** that includes a chipset attached memory controller **504**, and a chipset link **506**.

The system **500** also includes a processing unit package socket **508**, system memory slots **510**, and chipset attached memory slots **512**. The processing unit package socket **508** is designed to have a processing unit package, such as the processing unit package **102** of FIG. **1** or FIG. **4**, installed or inserted therein. In one or more implementations, the I/O expander **502** (including the chipset attached memory controller **504**), the chipset link **506**, one or more memory channels between the processing unit package socket **508** and the system memory slots **510**, and one or more memory channels between the I/O expander **502** and the chipset attached memory slots **512** are also referred to as a chipset of a motherboard or other printed circuit board. In one or more implementations, physically adjusting the system memory **104** involves removing one or more memory components from and/or inserting one or more memory components into the one or more system memory slots **512**.

(81) The I/O expander **502** is an I/O expander analogous to the I/O expander **106** of FIG. **1** or FIG. **4**. The chipset attached memory controller **504** is a chipset attached memory controller analogous to the chipset attached memory controller **116** of FIG. **1** or FIG. **4**. The chipset link **506** is analogous to the chipset link **110** of FIG. **1** or FIG. **4**.

(82) The system memory slots **510** include multiple (“x”) memory slots **510(1)**, **510(2)**, . . . , **510(x)**. The system memory slots **510** are designed to have system memory, such as the system memory **104** of FIG. **1** or FIG. **4**, installed or inserted therein. The system memory slots **510** are, for example, DIMM or SO-DIMM slots. Although multiple memory slots **510(1)**, **510(2)**, . . . , **510(x)** are illustrated, the system **500** optionally includes a single memory slot **510** or any other number of memory slots **510**.

(83) The chipset attached memory slots **512** include multiple (“y”) slots **512(1)**, **512(2)**, . . . , **512(y)**. The chipset attached memory slots **512** are designed to have chipset attached memory, such as the chipset attached memory **108** of FIG. **1** or FIG. **4**, installed or inserted therein. The chipset attached memory slots **512** are, for example, DIMM or SO-DIMM slots. Although multiple memory slots **512(1)**, **512(2)**, . . . , **512(y)** are illustrated, the system **500** optionally includes a single memory slot **512** or any other number of memory slots **512**. In one or more implementations, the number of chipset attached memory slots **512** is equal to or greater than the number of system memory slots **510** of the system memory **104**. In one or more implementations, if the number of chipset attached memory slots **512** is less than the number of system memory slots **510**, the system may determine portions of the contents of the system memory **104** which do not need to be transferred to the chipset attached memory. In one or more implementations, multiple chipset attached memories may be attached to the system, in which case the contents of the system memory **104** can be transferred to different ones of the multiple chipset attached memories.

(84) With a processing unit package installed or inserted in the processing unit package socket **508**, system memory installed or inserted in the system memory slots **510**, chipset attached memory installed or inserted in the chipset attached memory slots **512**, and additional I/O devices (e.g., a chipset attached nonvolatile memory and/or a chipset attached disk drive) optionally installed or otherwise coupled to the I/O expander **502**, the system **500** becomes the system **100** of FIG. **1** or a system having tiered memory options.

(85) It should be understood that many variations are possible based on the disclosure herein. Although features and elements are described above in particular combinations, each feature or element is usable alone without the other features and elements or in various combinations with or without other features and elements.

(86) The various functional units illustrated in the figures and/or described herein (including, where appropriate, the processing unit package **102**, the system memory **104**, the chipset attached memory **108**, the chipset link **110**, the processing unit **112**, the memory controller **114**, and the chipset attached memory controller **116**) are implemented in any of a variety of different manners such as hardware circuitry, software or firmware executing on a programmable processor, or any combination of two or more of hardware, software, and firmware. The methods provided are

implemented in any of a variety of devices, such as a general purpose computer, a processor, or a processor core. Suitable processors include, by way of example, a general purpose processor, a special purpose processor, a conventional processor, a digital signal processor (DSP), a graphics processing unit (GPU), a parallel accelerated processor, a plurality of microprocessors, one or more microprocessors in association with a DSP core, a controller, a microcontroller, Application Specific Integrated Circuits (ASICs), Field Programmable Gate Arrays (FPGAs) circuits, any other type of integrated circuit (IC), and/or a state machine.

(87) In one or more implementations, the methods and procedures provided herein are implemented in a computer program, software, or firmware incorporated in a non-transitory computer-readable storage medium for execution by a general purpose computer or a processor. Examples of non-transitory computer-readable storage mediums include a read only memory (ROM), a random access memory (RAM), a register, cache memory, semiconductor memory devices, magnetic media such as internal hard disks and removable disks, magneto-optical media, and optical media such as CD-ROM disks, and digital versatile disks (DVDs).

Claims

1. An apparatus comprising: a processing unit package comprising a processing unit and a memory controller; a system memory coupled to the processing unit package; a chipset attached memory coupled to the processing unit package via a chipset link; the memory controller configured to: provide a transfer request to the system memory, the transfer request causing contents maintained by the system memory to be transferred from the system memory to the chipset attached memory, wherein the system memory is disabled after the contents is transferred to the chipset attached memory; and provide an additional transfer request to cause the contents to be transferred from the chipset attached memory back to the system memory; and the processing unit configured to execute one or more tasks using the contents from the chipset attached memory.
2. The apparatus of claim 1, wherein power to the system memory is gated to disable the system memory.
3. The apparatus of claim 2, wherein gating the system memory enables one or more physical adjustments to be made to the system memory while the processing unit executes the one or more tasks using the contents from the chipset attached memory, wherein the one or more physical adjustments that can be made to the system memory include one or more of swapping a physical component of the system memory, removing a physical component of the physical memory, or replacing a physical component of the physical memory.
4. The apparatus of claim 3, wherein the physical component comprises a dual in-line memory module.
5. The apparatus of claim 1, wherein the memory controller is configured to provide the transfer request responsive to receiving an indication that one or more physical adjustments are going to be made to the system memory.
6. A method comprising: receiving an indication for making one or more physical adjustments to a system memory of a device, the system memory coupled to a processing unit of the device and a chipset attached memory coupled to the processing unit of the device via a chipset link; responsive to the indication, transferring, via the chipset link, contents of the system memory to the chipset attached memory; disabling the system memory after the contents is transferred to the chipset attached memory; while the system memory is disabled, operating the device using the contents from the chipset attached memory; and transferring, via the chipset link, the contents back from the chipset attached memory to the system memory after the one or more physical adjustments are made to the system memory.
7. The method of claim 6, wherein disabling the system memory comprises cutting power to the system memory while the one or more physical adjustments are made to adjust the system memory.

8. The method of claim 6, further comprising operating the device using the contents from the adjusted system memory after the contents are transferred back from the chipset attached memory.
 9. The method of claim 6, further comprising operating the device using the contents from the system memory before the contents are transferred to the chipset attached memory.
 10. The method of claim 6, wherein the one or more physical adjustments include one or more of swapping a physical component of the system memory, removing a physical component of the physical memory, or replacing a physical component of the physical memory.
 11. The method of claim 10, wherein the physical component comprises a physical memory module.
 12. The method of claim 11, wherein the physical memory module comprises a dual in-line memory module.
 13. The method of claim 6, further comprising training the adjusted physical memory before transferring the contents back from the chipset attached memory to the adjusted system memory.
 14. A system comprising: a processing unit package comprising a processing unit and a memory controller; system memory slots into which system memory is installable, the system memory slots coupled to the processing unit package; a chipset attached memory coupled to the processing unit package via a chipset link; the memory controller configured to cause contents stored in the system memory to be transferred to the chipset attached memory; and a processing unit configured to execute one or more tasks using the contents from the chipset attached memory while one or more system memory components are added or removed from one or more of the system memory slots.
 15. The system of claim 14, wherein the one or more system memory components include one or more physical memory modules.
 16. The system of claim 14, wherein the one or more system memory components include one or more dual in-line memory modules.
 17. The system of claim 14, wherein the memory controller is further configured to cause the contents to be transferred from the chipset attached memory back to the system memory.
 18. The apparatus of claim 1, further comprising an I/O expander that includes a chipset attached memory controller, wherein the chipset link couples the processing unit package to the I/O expander, and wherein the chipset attached memory is coupled to the I/O expander.
 19. The apparatus of claim 1, wherein the processing unit is configured to execute tasks using contents maintained by the chipset attached memory prior to transferring the contents to the chipset attached memory, wherein the contents maintained by the system memory is not maintained by the chipset attached memory prior to transferring the contents to the chipset attached memory.
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